

The Directed-Energy Debris Broom

Laser Ablation for Deorbiting Small Space Debris

Concept Study DB-CS-016 · Off-World Civilisation Toolkit · July 2026

Classification: Speculative engineering, constrained to known physics

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Abstract

Low Earth orbit is filling with shrapnel. Millions of fragments too small to track and too numerous to catch race around the planet at eight kilometres per second, each one able to cripple a satellite or hole a spacecraft, and each collision breeds more of them. You cannot grab a piece of debris a centimetre across moving faster than a rifle bullet — but you can push it with light. This study describes the DB-1 'Sweeper', a directed-energy system that fires precisely-timed laser pulses at catalogued debris fragments; each pulse boils a whisper of material off the fragment's surface, and the recoil acts as a tiny rocket, nudging the fragment's orbit down until it re-enters the atmosphere and burns up. No contact, no capture, one installation serving thousands of targets. We describe the ablation physics, the ground- and space-based architectures, the tracking that the system depends on, and a staged path from ground demonstration to operational clean-up. We also address, directly, the concept's unavoidable dual-use shadow, which is the real reason it belongs under international operation.

1. Introduction: The Shrapnel We Left Behind

Every satellite launched, every stage spent, every collision and explosion in orbit has left debris behind, and the debris does not fall quickly. In the busiest orbits, fragments can circle the Earth for decades or centuries, crossing the paths of working spacecraft at closing speeds where a fleck of metal carries the energy of a bullet. The tracked population numbers in the tens of thousands; the untracked centimetre-scale fragments, individually lethal to a satellite, number in the millions. Worse, each collision multiplies the hazard — a runaway cascade of collisions long ago warned of as a self-sustaining debris belt that could render whole orbits unusable (Kessler and Cour-Palais, 1978).

The small fragments are the intractable ones. Large derelicts can, in principle, be captured and removed one by one, but the centimetre-scale debris is too numerous to catch individually and too small to armour against — and it is exactly this population that drives the collision cascade. What is needed is a way to reach out and touch millions of tiny, fast-moving objects without flying to each one. Light can do that. Since the 1990s, laser ablation has been studied as a means of nudging small debris out of orbit from a distance, without ever making contact (Phipps et al., 1996), and the concept has matured into detailed ground- and space-based designs (Ebisuzaki et al., 2015). The Debris Broom is that idea built as an operational sweeper.

2. Concept Description

The DB-1 'Sweeper' is a directed-energy installation — on the ground, in orbit, or both — that removes small debris by laser ablation. It does not vaporise or shatter its targets; that would only create more fragments. Instead, working from a catalogue of tracked debris, it aims a pulsed laser at a selected

fragment and delivers a series of brief, intense pulses to its surface. Each pulse heats a microscopically thin layer of the fragment until a puff of material boils away, and the departing vapour, like the exhaust of a tiny rocket, pushes the fragment gently in the opposite direction.

Applied repeatedly and timed correctly, these nudges lower the fragment's orbit — specifically, they reduce the lowest point of its orbit until it dips deep enough into the upper atmosphere to be dragged down and burned up on re-entry. The fragment is not destroyed by the laser; it is gently persuaded to fall, and the atmosphere finishes the job. A single Sweeper, cycling through its target catalogue, can service thousands of fragments over its operating life, sweeping a lane of orbit clear without ever launching a chaser to any of them.

3. The Physics of Pushing with Light

The mechanism is laser ablation. When a sufficiently intense pulse of light strikes a solid surface, it heats the outermost layer so rapidly that the material does not merely warm but vaporises and blows off as a small jet of hot plasma. By conservation of momentum, that jet pushes back on the fragment it left — the ablation produces thrust. The pulse must be short and intense enough to ablate rather than simply heat, and aimed so that the resulting recoil points in the direction that lowers the orbit (Phipps et al., 1996). The individual impulse from one pulse is tiny, but debris orbits are precise and predictable, so many small, well-directed pushes accumulate into a decisive change in trajectory.

The decisive quantity is where the recoil points. To deorbit a fragment efficiently, the pulses are timed and directed to slow it at the right point in its orbit, lowering its perigee pass by pass until atmospheric drag takes over. This demands both a powerful, rapidly-pulsed laser and, above all, precise knowledge of each fragment's position and motion — which is why the Sweeper is inseparable from tracking. A ground-based system must also fire through the atmosphere, whose turbulence blurs the beam, requiring adaptive optics to keep the pulse tight on a small, distant, fast-moving target; a space-based system escapes the atmosphere but must carry its own power and cooling into orbit (Ebisuzaki et al., 2015). Both are engineering challenges, not physical impossibilities.

4. The Machine Itself

A ground-based Sweeper is, in essence, a large pulsed laser married to a precision telescope. The telescope tracks a catalogued fragment as it passes overhead; adaptive optics measure and correct for the atmosphere's distortion in real time; and the laser delivers its train of pulses during the brief window the fragment is in view and correctly positioned, nudging its orbit a little on each pass. Over many passes across many nights, the fragment's perigee is walked down until it re-enters. The installation never moves; the targets come to it, orbit after orbit.

A space-based Sweeper trades the atmosphere's blurring for the burden of carrying everything to orbit. Freed from atmospheric distortion, it can engage debris more directly and from closer range, but it must generate and cool the power its laser consumes and manage its own station-keeping. In either form, the true heart of the system is not the laser alone but its union with tracking and pointing: the Sweeper can only clean what it can precisely see, and its effectiveness is set as much by the debris catalogue and the pointing accuracy as by the beam itself.

5. Operations Concept

Phase one is ground demonstration: a laser-and-telescope facility that acquires a tracked fragment, corrects for the atmosphere, and imparts a measurable change to its orbit — proving the ablation nudge on a real target. Phase two scales to operational ground clean-up, systematically walking down the orbits of catalogued small fragments in the most congested altitudes, thinning the very population that drives the collision cascade. Phase three adds space-based Sweepers for debris beyond convenient ground reach and to protect specific high-value assets, building toward a maintained capability that keeps the busiest orbits swept clear. Throughout, the system's clean-up work is coordinated internationally, both to share its benefit and to govern its risks.

Parameter	Phase 1 (Ground demo)	Phase 3 (Operational)
Platform	1 ground laser+telescope	ground + space Sweepers
Target	one tracked fragment	catalogued small debris
Action	measurable orbit change	walk perigee down to re-entry
Key enabler	adaptive optics	tracking + pointing at scale
Effect on fragment	nudge (no shattering)	deorbit + atmospheric burn-up
Governance	national test	international operation

Table 1. Representative parameters for the staged programme.

6. Honest Difficulties

The technical difficulties are real: firing a tightly-focused, powerful pulsed beam onto a small, fast, distant fragment demands exquisite tracking and, from the ground, adaptive optics to defeat the atmosphere (Ebisuzaki et al., 2015); the ablation must nudge without fragmenting, since shattering a piece of debris would worsen the very problem being solved; and a space-based system's power and cooling are substantial burdens. But the concept's defining difficulty is not technical at all. A device that can nudge debris can, at higher power or closer range, damage a functioning satellite — the same beam that sweeps junk can dazzle or disable an asset. This dual-use character is unavoidable and must be met head-on: the Debris Broom is presented here strictly as a debris-removal system, and its legitimacy depends on transparent, cooperative, international operation with clear limits, precisely because the underlying capability could be misused. The physics is settled; the governance is the hard part, and it is the part that matters most.

7. Pathway to Reality

The ingredients are largely in hand. Laser ablation of debris has been studied and modelled for decades (Phipps et al., 1996); detailed ground- and space-based system concepts exist, and space agencies including JAXA and ESA have actively investigated debris-laser approaches (Ebisuzaki et al., 2015); adaptive optics is mature astronomical technology; and debris tracking improves continually. A ground demonstration imparting a measured nudge to a real fragment is a credible near-term step. The barrier to operational deployment is only partly technical; the larger obstacle is establishing the international framework under which an inherently dual-use capability can be trusted to sweep the commons without threatening what shares it. Solving the governance is as essential as building the laser.

8. Conclusion

Humanity has treated orbit as a place where things can simply be left, and the bill for that carelessness is now a sky full of shrapnel that threatens everything we fly. The Directed-Energy Debris Broom offers to sweep it clean with the gentlest of tools — a touch of light that persuades each fragment to fall of its own accord. Its physics is sound and its need is urgent. But its power to push a fragment is inseparable from a power to harm, and so the Broom, more than any other machine in this catalogue, is only as good as the trust built around it. Built openly and governed wisely, it could keep the orbits our future depends on open for everyone; that is a choice, and the choice matters as much as the machine.

References

Harvard referencing style (Cite Them Right).

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